

Model Assignment Solution

Title:

Calculation of GC Content in a DNA Sequence Using Python

1. Introduction

DNA sequences are composed of four nucleotides: Adenine (A), Thymine (T), Guanine (G), and Cytosine (C). The GC content of a DNA sequence refers to the percentage of guanine (G) and cytosine (C) bases present in the sequence. GC content is an important parameter in molecular biology because it influences DNA stability, melting temperature, and gene expression.

Computational tools help in calculating GC content quickly and accurately, especially for large genomic datasets.

2. Problem Statement

Write a Python program to calculate the GC content (%) of a given DNA sequence.

3. Algorithm

1. Start
2. Input the DNA sequence
3. Convert the sequence to uppercase
4. Count the number of G and C nucleotides
5. Count the total number of nucleotides
6. Calculate GC percentage using the formula:

$$GC\% = \frac{(G + C)}{\text{Total nucleotides}} \times 100$$
$$GC\% = \text{Total nucleotides} \times \frac{(G + C)}{100}$$

7. Display the GC content
8. Stop

4. Pseudocode

```
BEGIN
  INPUT DNA_sequence
  CONVERT DNA_sequence to uppercase
  SET G_count = count of 'G'
  SET C_count = count of 'C'
  SET total_length = length of DNA_sequence
  SET GC_content = ((G_count + C_count) / total_length) * 100
  PRINT GC_content
END
```

5. Flowchart (Text Representation)

```
Start
↓
Input DNA Sequence
↓
Convert to Uppercase
↓
Count G and C
↓
Calculate GC%
↓
Display Result
↓
Stop
```

(Student must draw this using standard flowchart symbols in the final submission.)

6. Python Program

```
# Program to calculate GC content of a DNA sequence

# Taking input from user
dna = input("Enter DNA sequence: ")

# Convert sequence to uppercase
dna = dna.upper()

# Count G and C
g_count = dna.count('G')
c_count = dna.count('C')

# Total length of sequence
total_length = len(dna)

# Avoid division by zero
if total_length > 0:
    gc_content = ((g_count + c_count) / total_length) * 100
    print("GC Content: {:.2f}%".format(gc_content))
else:
    print("Invalid DNA sequence.")
```

7. Sample Input and Output

Input:

ATGCGCATAAGC

Output:

GC Content: 50.00%

8. Result and Interpretation

The GC content of the given DNA sequence is 50%. This indicates that half of the nucleotides in the sequence are either guanine or cytosine. Higher GC content generally suggests greater thermal stability of the DNA molecule.

9. Conclusion

This assignment demonstrates how Python can be used to solve basic biological problems efficiently. Computational approaches are essential in biotechnology for analyzing sequence data and extracting meaningful biological insights.